HEAVY METAL TESTED & CERTIFIED · CERTIFICATION STANDARD

Infant formula (powder) — HMTc certification standard

Institute of Contaminant Standards — Heavy Metal Tested & Certified program

Version 1.0 · Native basis: as_sold · 8/8 analytes with a published limit

STANDARD

infant-formula-powder, version 1.0. HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

PROVENANCE

Compiled from data/evidence/regulatory_limits.csv. Each limit is a lookup against sovereign law, not an occurrence computation.

TAMPER-EVIDENT SEAL

The SHA-256 of this exact PDF is sealed by an RFC-3161 trusted timestamp. The token (standard.pdf.tsr) and the manifest (manifest.json) accompany this file; the date of record is the timestamping authority's, not the program's.

Abstract

This is the frozen, version-pinned Heavy Metal Tested & Certified standard for infant formula (powder). Each per-analyte certification limit is the strictest heavy-metal maximum level set by any government in the world for this product class, expressed in the product's native basis (as_sold) and derived by lookup against sovereign law under HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The per-jurisdiction derivation below is the complete "show your work". This snapshot is immutable: any material change to a limit mints a new version with its own timestamped snapshot.

1 | Certification limits

All values in ppb (µg/kg), as_sold basis. A brand meeting every published value on a like-for-like basis qualifies for the mark. Tier 1 toxics (Pb, Cd, iAs, mercury) carry a zero-tolerance posture; Tier 2 metals (Ni, Al, Sn, total Cr) carry the program's transitional allowance. Where a limit is set in another basis by the binding jurisdiction it is converted into the native basis and the conversion is named in the "Basis derivation" column.

Cow-milk based

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	20	European Union	same basis	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	20	European Union	same basis	regulatory-alignment
Cadmium (Cd)	Tier 1	10	European Union	same basis	regulatory-alignment
Mercury (tHg)	Tier 1	15	Israel	same basis	regulatory-alignment
Nickel (Ni)	Tier 2	250	European Union	same basis	regulatory-alignment

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Aluminium (Al)	Tier 2	4000	Australia / NZ	×8 as-consumed→as-sold	regulatory-alignment
Tin (Sn)	Tier 2	5000	India	same basis	regulatory-alignment
Total chromium (Cr)	Tier 2	2000	China	same basis	regulatory-alignment

Soy based

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	20	European Union	same basis	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	20	European Union	same basis	regulatory-alignment
Cadmium (Cd)	Tier 1	10	Israel	same basis	regulatory-alignment
Mercury (tHg)	Tier 1	15	Israel	same basis	regulatory-alignment
Nickel (Ni)	Tier 2	400	European Union	same basis	regulatory-alignment
Aluminium (Al)	Tier 2	8000	Australia / NZ	×8 as-consumed→as-sold	regulatory-alignment
Tin (Sn)	Tier 2	5000	India	same basis	regulatory-alignment
Total chromium (Cr)	Tier 2	2000	China	same basis	regulatory-alignment

2 | Per-jurisdiction derivation

The binding value for each analyte is the lowest sovereign maximum level across the jurisdictions on record, in the native basis. Rows marked "converted" were set by the jurisdiction in the other basis and carried across the reconstitution ratio; rows marked "native" were already in this basis. Every value is keyed to its record in the sovereign register.

Cow-milk based

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	European Union-(binding)	20	native	eu2023_pb_formula_powder_20
—	Fiji	20	native	fj_fsr2009_pb_006_infant_formula_20
—	Israel	50	native	il_fcs2016_pb_15_1_infant_formula_powder_50
—	United Kingdom	50	native	uk_pb_infant-formulae-and-follow-on--formulae-m_50
—	Australia / NZ	80	converted	aunz_sch19_pb_formula_10
—	Brazil	80	converted	br_pb_pediatric-formula-for-enteral--nutrition_10
—	Canada	80	converted	ca_pb_infant-formula-all-types-two-former--limi_10

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—	China	80	native	cn_pb_formula-food-for-infants-and-young--child_80
—	Codex Alimentarius	80	converted	codex_pb_infant-formula-formula-for--special-medic_10
—	Gulf (GSO)	80	converted	gso193_2021_pb_infant_formula_formula_for_-special_medical_purposes_intended_for_-infants_10
—	HK	80	converted	hk_cap132v_pb_infant-formula-and-follow--up-formula_10
—	Singapore	80	converted	sg_foodregs_2026_infant_formula_as_-consumed_pb_10
—	Thailand	80	converted	th_moph414_2020_pb_formula_as_consumed_10
—	Colombia	160	converted	co_res4506_2013_pb_formula_ready_20
—	India	200	native	in_pb_infant-milk-substitute-and-infant--foods_200
iAs	European Union- (binding)	20	native	eu2023_ias_formula_powder_20
—	Israel	50	native	il_fcs2016_ias_14_1_infant_and_toddler_-formulas_powder_50
—	Brazil	160	converted	br_ias_pediatric-formula-for-enteral--nutrition_20
Cd	European Union- (binding)	10	native	eu2023_cd_formula_cow_powder_10
—	Gulf (GSO)	10	native	gso193_2021_cd_powdered_formulae_-manufactured_from_cows_milk_proteins_or_-protein_hydrol_10
—	Israel	10	native	il_fcs2016_cd_14_1_infant_and_toddler_-formulas_powder_10
—	United Kingdom	10	native	uk_cd_infant-formulae-and-follow-on--formulae-p_10
—	Brazil	80	converted	br_cd_pediatric-formula-for-enteral--nutrition_10
—	India	100	native	in_cd_infant-milk-substitute-and-infant--foods_100
tHg	Israel (binding)	15	native	il_fcs2016_thg_14_1_infant_and_toddler_-formulas_powder_15
Ni	European Union- (binding)	250	native	eu2024_ni_formula_powder_250
Al	Australia / NZ (binding)	4000	converted	aunz_sch19_al_formula_nonsoy_500
Sn	India (binding)	5000	native	in_sn_infant-milk-substitute-and-infant--foods_5000
Cr	China (binding)	2000	native	cn_crtotal_milk-powder-and-modified-milk--powder-gen_2000

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Soy based

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	European Union- (binding)	20	native	eu2023_pb_formula_powder_20
—	Fiji	20	native	fj_fsr2009_pb_006_infant_formula_20
—	Israel	50	native	il_fcs2016_pb_15_1_infant_formula_powder_50
—	United Kingdom	50	native	uk_pb_infant-formulae-and-follow-on--formulae-m_50
—	Australia / NZ	80	converted	aunz_sch19_pb_formula_10
—	Brazil	80	converted	br_pb_pediatric-formula-for-enteral--nutrition_10
—	Canada	80	converted	ca_pb_infant-formula-all-types-two-former--limi_10
—	China	80	native	cn_pb_formula-food-for-infants-and-young--child_80
—	Codex Alimentarius	80	converted	codex_pb_infant-formula-formula-for--special-medic_10
—	Gulf (GSO)	80	converted	gso193_2021_pb_infant_formula_formula_for_-special_medical_purposes_intended_for_-infants_10
—	HK	80	converted	hk_cap132v_pb_infant-formula-and-follow--up-formula_10
—	Singapore	80	converted	sg_foodregs_2026_infant_formula_as_-consumed_pb_10
—	Thailand	80	converted	th_moph414_2020_pb_formula_as_consumed_10
—	Colombia	160	converted	co_res4506_2013_pb_formula_ready_20
—	India	200	native	in_pb_infant-milk-substitute-and-infant--foods_200
iAs	European Union- (binding)	20	native	eu2023_ias_formula_powder_20
—	Israel	50	native	il_fcs2016_ias_14_1_infant_and_toddler_-formulas_powder_50
—	Brazil	160	converted	br_ias_pediatric-formula-for-enteral--nutrition_20
Cd	Israel (binding)	10	native	il_fcs2016_cd_14_1_infant_and_toddler_-formulas_powder_10
—	European Union	20	native	eu2023_cd_formula_soy_powder_20
—	Gulf (GSO)	20	native	gso193_2021_cd_powdered_formulae_-manufactured_from_soya_protein_isolates_-alone_or_in_a__20

—	United Kingdom	20	native	uk_cd_infant-formulae-and-follow-on--formulae-p_20
—	Brazil	80	converted	br_cd_pediatric-formula-for-enteral--nutrition_10
—	India	100	native	in_cd_infant-milk-substitute-and-infant--foods_100
tHg	Israel (binding)	15	native	il_fcs2016_thg_14_1_infant_and_toddler_-formulas_powder_15
Ni	European Union- (binding)	400	native	eu2024_ni_formula_soy_powder_400
Al	Australia / NZ (binding)	8000	converted	aunz_sch19_al_formula_soy_1000
Sn	India (binding)	5000	native	in_sn_infant-milk-substitute-and-infant--foods_5000
Cr	China (binding)	2000	native	cn_crtotal_milk-powder-and-modified-milk--powder-gen_2000

3 | Method and provenance

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The default standard for any (product, analyte) is the strictest maximum level set by a credible government regulator, converted to the row's native basis. Where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. Species limits are bounded by their parent total (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ total Cr); read-across limits (for example aluminium carried from an as-consumed limit to an as-sold basis across the reconstitution ratio) are disclosed as read-across and never presented as direct governance.

The occurrence and toxicology corpus that contextualises these limits is maintained independently as the Heavy Metal Index literature baseline (heavymetalindex.com). This certification standard cites that baseline one way; it does not use the baseline's contents as the basis for any government-limit value stated here.

4 | Version and tamper-evidence

This is version 1.0 of the Infant formula (powder) standard. A certification number becomes final when its version carries a resolvable DataCite DOI; a material change to any limit mints a new version with its own frozen snapshot, so a superseded number is never silently overwritten. Tamper-evidence does not depend on trusting the program: the SHA-256 of this exact PDF was submitted to an RFC-3161 trusted timestamping authority (https://freetsa.org/tsr), which returned a signed token binding that hash to a time it — not the program — attests. The token (standard.pdf.tsr), the authority's CA chain (tsa-cacert.pem), and the manifest (manifest.json) travel with this file.

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